

HealthConnect — Week 8 Project Summary

HealthConnect Experience Lab — Data Analytics Track

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1. Project Overview

The HealthConnect Experience Lab focused on answering the project question:

“How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?”

The Data Analytics track focused on analysing appointment data to understand patterns associated with missed appointments, validate important findings, develop decision-support visuals, and translate the results into practical business insights and recommendations.

The Week 8 work continued the findings and outputs developed during Weeks 4–7 rather than introducing a new dataset or separate project.

2. Data and Analytical Approach

The analysis used the HealthConnect appointment dataset containing 5,000 appointment records and 18 fields.

The Data Analytics workflow included:

1. Data quality assessment and preparation
2. Exploratory data analysis
3. KPI development
4. Segment analysis
5. Statistical validation
6. Cross-track validation
7. Power BI dashboard development
8. Dashboard testing and refinement
9. Business insight development
10. Final integration and presentation preparation

The analysis focused on factors including booking lead time, previous No-Show history, distance to clinic, reminder status and channel, appointment type, gender, and combined appointment segments.

3. Final KPIs

The final dashboard contains five validated KPI cards:

- Overall No-Show Rate — 48.5%
- No Reminder — 51.4%
- Previous No-Show — 55.4%
- 3+ Week Lead Time — 56.9%
- Highest Observed Segment — 64.1%

The Highest Observed Segment is defined as:

3+ weeks (≥ 22 days) + Previous No-Show | All outcomes included

This definition is explicitly documented to distinguish it from the separately defined Data Science segment used during cross-track validation.

4. Key Validated Findings

Booking Lead Time

No-Show Rates increased across the booking lead-time groups:

- Same week (0–7 days): 27.8%
- 2–3 weeks (8–21 days): 37.2%
- 3+ weeks (22–60 days): 56.9%

The difference between the shortest and longest lead-time groups was 29.1 percentage points.

Statistical testing identified an association between booking lead time and appointment outcome:

$\chi^2 = 280.09$, $p < 0.0001$, Cramér's $V = 0.167$.

Previous No-Show History

Appointments involving patients with at least one previous No-Show had a higher observed No-Show Rate:

- No previous No-Show: 43.5%
- Previous No-Show: 55.4%

This represents an 11.9 percentage-point difference.

Statistical testing identified an association between previous No-Show history and appointment outcome:

$\chi^2 = 69.86$, $p < 0.0001$, Cramér's $V = 0.118$.

Combined Lead Time × Previous No-Show

Combining booking lead time and previous No-Show history produced clearer separation between observed appointment segments.

The results were:

- Same week + No Previous No-Show: 21.8%
- Same week + Previous No-Show: 36.2%
- 2–3 weeks + No Previous No-Show: 34.1%
- 2–3 weeks + Previous No-Show: 41.9%
- 3+ weeks + No Previous No-Show: 51.7%
- 3+ weeks + Previous No-Show: 64.1%

The combined segmentation had:

$\chi^2 = 358.52$, $p < 0.0001$, Cramér's $V = 0.189$.

This was the strongest association among the tested segmentation approaches.

These results describe observed associations and should not be interpreted as proof of causation or predictive improvement.

5. Final Dashboard

The final Power BI dashboard contains:

- 5 KPI cards
- 6 comparison visuals
- 2 slicers: Appointment Type and Gender

The dashboard presents:

1. No-Show Rate by Booking Lead Time
2. No-Show Rate by Previous No-Show History
3. No-Show Rate by Distance to Clinic
4. No-Show Rate by Reminder Status
5. No-Show Rate by Reminder Channel
6. Lead Time × Previous No-Show History

The dashboard was tested against the underlying dataset, including KPI values, visual calculations, slicer behaviour, combined analysis, and lead-time ordering.

A dedicated Lead Time Band Sort column was created during Week 7 to correct the lead-time category ordering, and the affected visuals were successfully re-tested.

6. Cross-Track Contribution

The Data Analytics track collaborated with Data Science — Donald Nwachukwu — to validate an important segment calculation.

Data Science used the following segment definition:

- `booking_lead_days > 30`
- `previous_no_shows >= 1`

The original Data Science calculation excluded Cancelled appointments:

$689 \text{ No-Shows} \div 977 \text{ No-Show} + \text{Attended} = 70.52\%$.

Data Analytics independently reproduced the calculation and also tested the same segment with Cancelled appointments included:

- Cancelled excluded: 70.52%
- Cancelled included: 67.88%
- Total appointments in the segment: 1,015

The testing identified two differences between the original analyses:

1. Different lead-time thresholds.
2. Different denominator definitions.

The exact Data Science segment was successfully reproduced, confirming the calculation under its stated definition and improving consistency between the two tracks.

Final End-to-End Integration

The HealthConnect solution combines the work of multiple tracks to address missed appointments and improve the patient support experience. Data Analytics provides validated KPIs, analytical findings, dashboard insights, and business recommendations based on the appointment data. Data Science contributes predictive modelling, while ML Engineering supports the technical workflow, GenAI supports the healthcare administrative experience, and Project Management coordinates the integration of the different workstreams.

The validated Data Analytics findings, particularly the relationship between booking lead time and No-Show Rates and the role of previous No-Show history, provide an evidence base for decision support and contribute to the interpretation of the wider HealthConnect solution.

The overall workflow can be summarized as:

Problem → Data → Analytics → Predictive Modelling → ML Engineering → GenAI → Project Management → Final HealthConnect Solution

The solution remains subject to limitations including the synthetic nature of the dataset, observational analysis, subgroup size limitations, and the need for continued validation of predictive performance and real-world effectiveness.

7. Business Insights

The analysis identified several important areas for decision support.

Longer Booking Lead Times

Longer booking lead times showed higher observed No-Show Rates. This suggests that HealthConnect could consider additional appointment support for appointments booked well in advance.

Previous No-Show History

Previous No-Show history was associated with a higher observed No-Show Rate. This factor can therefore be considered when developing appointment-support strategies.

Combined Segmentation

Combining booking lead time with previous No-Show history provided greater separation between observed appointment segments and can support more focused decision-making.

Reminder Patterns

Differences were observed across reminder status and reminder channels. However, these results are observational and do not establish that a particular reminder method causes lower No-Show Rates.

Distance

Distance showed an observed relationship with No-Show Rate, with the 15km+ group having a higher observed rate than the shorter-distance groups. Distance is therefore retained as a supporting analytical factor.

8. Business Recommendations

Based on the validated findings, the Data Analytics track recommends that HealthConnect:

1. Prioritize longer-lead-time appointments for additional appointment support.
2. Consider previous No-Show history when identifying appointments that may require additional support.
3. Use the combined lead-time and previous-history segmentation as a decision-support approach.

4. Continue monitoring reminder patterns and validate their effectiveness through further testing.
5. Use the Power BI dashboard as an ongoing decision-support tool for monitoring appointment patterns.

These recommendations are based on observed relationships in the available dataset and should be validated further before being treated as evidence of real-world impact.

9. Testing, Refinement and Validation

The Data Analytics outputs underwent testing and refinement during Weeks 7 and 8.

The testing confirmed:

- KPI calculations
- Comparison visual values
- Appointment Type slicer
- Gender slicer
- Combined Lead Time × Previous No-Show analysis
- Lead Time Band ordering
- Cross-track segment calculation

The main dashboard issue identified during testing was the Lead Time Band ordering. This was resolved using a dedicated sort column and successfully re-tested.

No confirmed KPI calculation errors remained after refinement and re-testing.

10. Limitations

The main limitations of the final analysis are:

- The dataset is synthetic and may not represent real-world patient behaviour.
- Previous appointment and No-Show history fields may not represent fully realistic patient histories.
- Some variables contain missing values.
- The analysis identifies associations rather than proving causation.
- Statistical significance does not by itself demonstrate predictive performance.
- Different segment definitions and denominator rules can produce different reported rates.
- Some subgroup analyses contain relatively small sample sizes.
- Reminder analysis remains observational.
- Dashboard testing covered the tested KPI, visual, slicer, and configuration scenarios and does not guarantee behaviour under every possible future data or filter state.

11. Final Contribution to HealthConnect

The Data Analytics track contributes a validated evidence base for understanding appointment No-Show patterns within the available HealthConnect dataset.

The final contribution includes:

- A validated Power BI decision-support dashboard
- Five final KPIs
- Statistically validated analytical findings
- Business insights
- Evidence-based recommendations
- Testing and refinement evidence
- Cross-track validation with Data Science
- Documented limitations
- Final presentation-ready analytical outputs

The overall contribution follows the workflow:

HealthConnect Data → Analysis → Validation → Dashboard → Insights → Recommendations → Cross-Track Integration

12. Final Conclusion

The HealthConnect Data Analytics work identified booking lead time and previous No-Show history as important observed factors associated with appointment outcomes in the synthetic dataset.

Combining these factors provided stronger separation between observed appointment segments, while the Power BI dashboard translated the validated findings into an interactive decision-support tool.

The Data Analytics component has been tested, refined, validated, and prepared for final integration and presentation.

The findings can support the wider HealthConnect solution while remaining subject to the documented limitations of the dataset and analytical approach.